

CHEMICAL FAMILIES

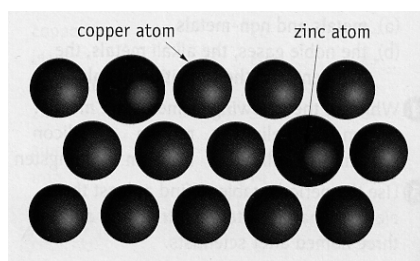
METALS

Properties

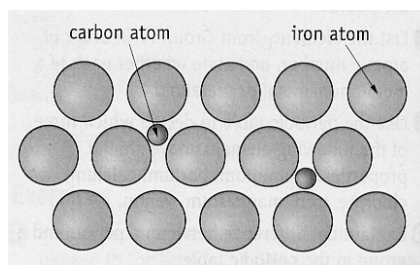
- Good conductors of heat and electricity.
- *Malleable* - can be hammered into sheets.
- *Ductile* - can be drawn/pulled into wires.
- *Metallic lustre* - shiny when polished.

Alloys

- Mixture of two or more metals.
e.g. Silver coins are 75% Cu and 25% Ni.
- Can have atoms of the same size.



- Can have atoms of different sizes.



Reactive Metals

- Group I metals are very reactive (have 1 electron in the outer shell).
- Called *alkali metals*.
e.g. Na in water can explode.
- Group II metals are not as reactive (have 2 electrons in the outer shell).

Transition Metals

- In the middle of the periodic table.
- Hard with high melting points.
- Used to form many alloys.

- Form coloured compounds with non-metals.

e.g. CuSO_4 is blue.

- Cu^{2+} ions are blue.
 Fe^{2+} ions are green.

NON-METALS

Properties

- Found on right hand side of the periodic table.
- Most are gases at room temperature.

Group VII

- Called the *halogens* - very reactive as they have room for one extra electron in their outer shell.
- Form *salts* when combined with metals.

Group VIII

- Are *monatomic* - have one atom only.
- The other non-metal gases are *diatomic* - two atoms together.

e.g. N_2 , O_2 , Cl_2 , F_2 .

Allotropes

- Different forms of the same element.

e.g. Carbon has diamond (hard) and graphite (soft).

